



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
Lipa Campus

A. Tanco Drive, Marawoy Lipa City, Batangas, Philippines 4217
Tel Nos.: (+63 43) 779-8400 local 3106

E-mail Address: ovcaa.lipa@g.batstate-u.edu.ph | Website Address: <http://www.batstate-u.edu.ph>

Office of the Vice Chancellor for Academic Affairs

Title

Lab Equipment Borrowing and Tracking System

Description

The Lab Equipment Borrowing and Tracking System is designed to provide an efficient and reliable system for managing and monitoring laboratory equipment within the school. By integrating RFID technology, the system automates the borrowing and returning process, reducing manual errors and saving time for both students and staff. Each piece of equipment will be tagged with an RFID sticker, and users will utilize their RFID-enabled IDs to borrow or return items. This eliminates the need for manual entry, ensures accurate records, and enhances accountability in the use of school laboratory resources. The system ultimately aims to streamline equipment management, promote proper utilization, and prevent loss or misplacement of laboratory assets.

General Objectives

- To develop a digital system that automates the borrowing and returning of laboratory equipment using RFID technology.
- To establish a reliable tracking mechanism that records the real-time status and history of borrowed equipment.
- To minimize manual processes and human error in logging laboratory equipment transactions.
- To promote accountability among students and staff in handling laboratory equipment.
- To support laboratory administrators in managing inventory through accurate and accessible records.

Integration Analysis

The system integrates three major components:

- Hardware Integration - Uses RFID/IoT tags, sensors, and microcontrollers to track users and equipment.
- Database Server - Centralized database stores borrow/return logs and equipment status.
- Network Integration – Devices transmit data securely to the central system.

*Data flow is analyzed to ensure seamless communication between the hardware and software layers.
Integration testing is performed to verify interoperability.*

Agile Methodology

Sprint Planning:

The system development is divided into multiple sprints, each focusing on specific core features.

- **Sprint 1:** Implementation of user authentication through RFID-based login and sign-up features.
- **Sprint 2:** Development of the landing page and inventory management interface.
- **Sprint 3:** Integration of the borrowing and returning process with real-time equipment tracking using RFID.
- **Sprint 4:** Creation of the dashboard and report generation modules.

Daily Stand-ups: Short daily check-ins to share progress, identify challenges, and set immediate goals for the next development tasks.

Sprint Review: Demonstration of completed features such as RFID authentication, equipment tracking functionality, database updates, and dashboard analytics to the instructor or client for feedback

Sprint Retrospective: The team reviews the sprint process, discusses challenges encountered, and identifies actionable ways to improve teamwork and efficiency for the next cycle.



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User and System Requirements (Hardware and Software)

For User:

- Must be able to borrow and return laboratory equipment quickly and easily.
- Must have a unique RFID ID for identification during transactions.
- Should be able to check their borrowing history and status.

Laboratory Staff:

- Must be able to scan equipment to record borrowing and returning.
- Should be able to verify the borrower's identity via RFID.
- Must have access to a dashboard to view current availability of items.
- Should be able to generate reports on borrowed, returned, and missing equipment.

For Hardware Requirements:

- **Microcontroller:** ESP32 – Serves as the main processing unit, handling RFID scans and communicating with the database/server through Wi-Fi.
- **RFID Reader:** RC522 Module – Used for scanning RFID cards/tags assigned to users and equipment.
- **RFID Tags/Stickers:** Each tag contains a unique ID used for identifying students and equipment during borrowing or returning.
- **LED Indicators:** Provide visual feedback (e.g., green for success, red for error).
- **Breadboard & Jumper Wires:** Used for circuit prototyping and component connections.
- **Power Supply/USB Cable:** Provides power to the ESP32 and peripheral components during testing and operation.

For Software Requirements:

- **Operating System:** Windows
- **Programming Language:** Java
- **Database:** MySQL or PostgreSQL
- **Development Environment:** Visual Studio Code / Eclipse
- **Firmware:** Arduino IDE or PlatformIO



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System Model / Design Diagram

Description:

The system follows a **3-Tier Architecture** adapted for an application setup:

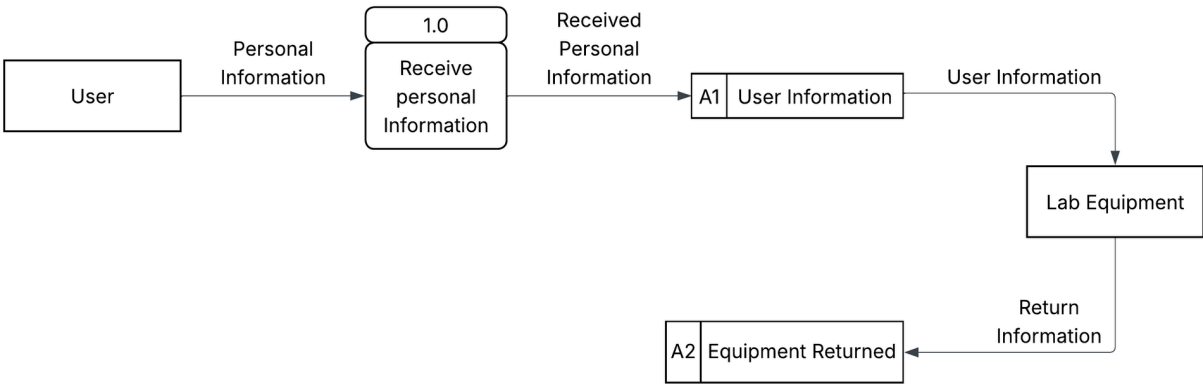
Presentation Layer – Provides the interface for students and administrators to borrow/return items, track equipment status, and view logs.

Application Layer – Processes input from the RFID hardware, verifies transactions, and enforces/returning rules.

Data Layer – Centralized database storing student information, equipment inventory, and transaction logs.

Data Flow Diagram

Borrowing Process



Returning Process

